

Digital Twin : An innovative antenna optimization technology

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Abstract— A Digital Twin is an innovative technology that links the real and virtual world by collecting real-time data from the installed sensors. This technology benefits various sectors such as astronomy, automotive, and manufacturing. The goal of this paper is to review the existing literature and explore the role of Digital Twin and its applications for Industry 4.0 in general, with a particular focus on antennas optimizations, in the end we will focus on active digital twin as result of AI integration.

Keywords— Digital Twin, antennas optimizations, machine learning, Industry 4.0, AI

I. INTRODUCTION

With the rise of global connectivity, Industry 4.0 has led many changes and among these changes the introduction of Digital Twin. Digital Twin has made the role of antennas more critical than ever. Recent efforts by several researchers around the world to provide seamless w

wireless connectivity in 5G and beyond have led to the inclusion of AI in antenna optimization with the goal of improving key parameters such as enhancing gain, efficiency, bandwidth radiation pattern and more. In this paper, we review and explores the Digital Twin, a technology that has grown in popularity and which does not stand still.

We will begin by reviewing the history of the Digital Twin and its initial definition, to be followed by a close examination of its goals in terms of optimization. The increasingly interlinked integration of artificial intelligence into Digital Twin marks the beginning of an exciting new era for the principles of antenna optimization. This article therefore permits to take a look on the basics of Digital Twin, the key role of AI in its development and the challenges and future directions on this technology.

II. DIGITAL TWIN

A. History of Digital Twin Technology

The Digital Twin concept earned widespread recognition in 2002 following the presentation at Challenge Advisory by Dr. Michael Grieves a professor at the University of Michigan[1]. But it was Initially introduced in the book the Mirror Worlds, published in around 1991 by David Gelernter. And finally, the

term “Digital Twin” was coined in 2010 in a NASA report by John Vickers. Digital Twin had already been put into use much earlier in the 1960s when NASA launched Apollo 13 spacecraft up into the space and deployed a simulator on the ground in Houston to follow along on the mission. Since 2010 until now, and with the emergence of smart manufacturing, Digital Twin evolved into a more robust technology and saw widespread use across industries. The Digital Twin concept has undergone tremendous evolution since its emergence.

Table 1 : Digital Twin evolution

Year	Description
2002 (First Introduction)	Michael Grieves introduced the concept of a “Digital Twin” during his discourse on Product Lifecycle Management (PLM).
2003 (Early Development)	Framling and his team proposed a system where each product has a Digital Twin, accurately mirroring the product’s status and information.
2010 (Practical Application)	NASA began creating virtual counterparts for its space assets.
2012 (Concept Expansion)	The use of Digital Twins expanded to disciplines like aircraft maintenance and air force product management.
2014 (Formal Definition)	Grieves defined the concept as a system involving physical entities, their virtual equivalents, and the data exchange bridging physical and virtual realms.
2024 (Recent Applications)	Digital Twin technology became widely deployed across sectors such as manufacturing and construction.

B. Digital Twin Optimization Goals

Digital Twin Optimization allows to tackle various challenges and provides significant benefits:

- **Enhanced Design Accuracy and real-Time Monitoring [2]:** Digital Twin plays a key role in enhancing accuracy, continuous monitoring and providing high efficiency for the antenna design before physical prototypes are created. An experiment has been conducted in this regard by comparing two sets of data Real-World data and Digital Replica data, the objective was to evaluate the proposed Digital Twin aided communications. The experiment showed that Digital Twin can be used effectively to predict optimal beams.
- **Optimized Performance[3] :** Antenna Digital Twin offer virtual representation of the antenna, and assist in optimizing parameters such as gain, efficiency, and bandwidth, Guaranteeing improved overall performance. An experiment was performed, where the authors used an RF dataset collected from an autonomous car [3].
- **Cost Efficiency:** Reducing material cost by minimizing the necessity of multiple physical prototypes.
- **Customization and Adaptability:** The ability to adjust in design and allow quick customization to meet individual user needs.
- **Decision Making:** with Digital Twin the impact can be predicted, developers and architects are able to make better-informed decisions

C. Difference between Digital Twin and Simulation

If Digital Twin is a simulation of a physical object, then, what makes the difference between Digital Twin and simulation? Since the purpose of both of them is modeling, then, why do we put so much effort into creating a Digital Twin via different tools instead of using just the traditional one? Why all this focus on Digital Twin as an essential driver for industry 4.0 transformation? In the case of our study, for example, since the CST can yield good 3D electromagnetic simulation with modeling of antenna radiation and all the parameters, then why create a twin for our antenna? The simulation and Digital Twin share the same overarching concept when viewed from a global perspective, though the difference between them is really multifaceted. First, regarding the simulation, it's a model designed in a virtual environment to mimic a system or process in specific parameters and under conditions of a specific environment while using mathematical framework tools and relations. In the majority of cases, we don't have control of this environment, , this leads to study the process from just one perspective and subsequently the digital representation is limited on a particular process. There are a lot of tools of simulation dedicated to specific uses, the input is a predefined data fix and operates in a static time, consequently, the output is generated based on fixed data. In turn, the simulation is a static representation limited to specific conditions.

On the other hand, Digital Twin is a simulation with real-time data, which means that the input is available immediately or with minimal delay. Consequently, the data here is continuously updated and reflects different scenarios and different conditions. As a result, this digital representation can predict and optimize processes, especially with the added benefits of the possibility of AI integration. Is that why the Digital Twin is an active representation where the data can be altered easily?

While the simulation is limited in one process, the Digital Twin allows us to study multiple simulations simultaneously. Also, Digital Twin is known for the high level of precision and accuracy in the following triangle.

The RAP model illustrated in Fig. 1 illustrates the main characteristics of Digital Twin.

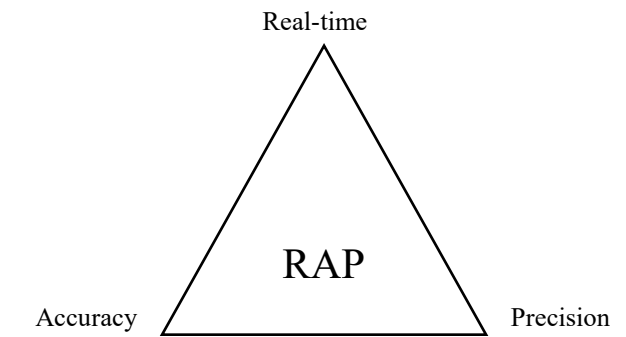


Fig. 1: RAP model

To facilitate understanding the differences between simulation and Digital Twin, Table 2 summarizes the key elements:

Table 2: Differences between Simulation and Digital Twin

Criteria	Simulation	Digital Twin
Connection to the reality	No	Yes , in real time
Type of simulation	Static	Active
Operation	Specific scenarios	Continuous
Objective	Analysis and design	Monitoring , optimization , prediction
integration with AI	No	Yes

D. Digital Model vs. Digital Shadow vs. Digital Twin

The distinction between a Digital Model, a Digital Shadow, and a Digital Twin is unclear in many cases. Further, in what follows, we attempt to distinguish these three terms. The main difference between Digital Twin, Digital Shadows, and Digital Model is the nature and direction of the data flow between the physical and virtual systems (See Fig. 2).

- **Digital Model:** A Digital Model is a virtual representation of a physical object. There is no automatic data exchange between the Digital Model and the physical object, meaning alterations to one do not influence the other.
- **Digital Shadow:** In this scenario, data flows from the physical object to the Digital Model. This continuous stream of information keeps the Digital Model updated with new insights from the real world, accurately mirroring the current state of its physical counterpart.
- **Digital Twin:** In a Digital Twin, data flows bidirectionally between the physical object and its digital counterpart. Any change to the physical object is automatically mirrored in the Digital Model, and vice versa. This integration guarantees that the DT's is permanently synchronized with the current physical object's status, making real-time monitoring and control possible.

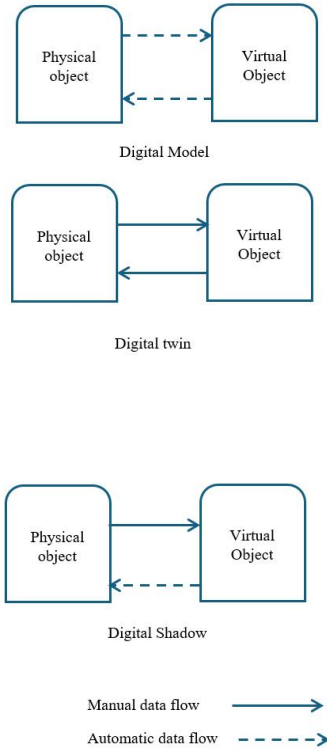


Fig. 2: Differences Between Digital Model, Digital Shadow and Digital Twin.

E. Key Requirements for Digital Twin

The fundamental requirements essential to the development of Digital Twin are listed below:

Model: Describes a virtual replica of physical objects used in the context of Digital Twin technology to predict and simulate the physical properties in real time.

Framework: The terminology represents the core software which involves the development of Digital Twin. The Framework is empowered to alter numerous data forms counting, data inputs, contextual information and sensor metrics.

Application: To ensure the digital twin meets its objectives in achieving business objectives, an application is necessary.

By working within the framework this application can generate data and relevant information which might result in concrete business enhancements.

Review of current Digital Twin platforms
In Annex 1, you will find a table illustrating the review of current Digital Twin platforms and a comparison between them based on various criteria. The table includes an analysis of platforms such as Azure Digital Twins, Amazon IoT Twin Maker, Oracle IoT Digital Twin Framework, and Eclipse Ditto

F. Review of Digital Twin Across Application Domains

Our literature review, conducted using Dimensions.ai, focused on papers from IEEE Access and articles published between 2020 and 2024 with the keyword "Digital Twin". We Selected 11,773 papers, excluding those from predatory journals. The majority of these papers are in the manufacturing domain, accounting for 35.1% of the total, highlighting their prominent role in the Digital Twin field. Healthcare follows with 19.2%, reflecting its growing adoption of Digital Twin technologies in diagnostics and treatment. Other significant domains include energy (12.6%), aerospace (8.4%), civil infrastructure (7.5%), and smart cities (7.1%). Figure. 3 presents the arrangement of selected papers by Application Domains on Digital Twin, proving the importance of Digital Twin manufacturing and the increasing of Digital Twin in many industries.

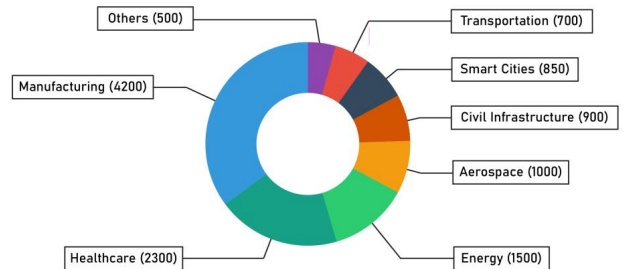


Fig. 3: Distribution of Selected Papers by Application Domains on Digital Twin

III. ANTENNA OPTIMIZATION

A. Definition of Machine Learning

ML, the abbreviation of machine learning, is a branch of artificial intelligence that allows systems to learn from data provided and recognize patterns, it also makes decisions or predictions based on this data with minimal guidance from humans.

B. Antenna Optimization

Optimization of antenna is a critical and fundamental field in telecommunication, considering its capacity to improve the communications performance. This concept centers on essential elements that aim to ensure the best operation of antenna, which reflect various parameters like gain and directivity.

The main goal is to improve the signals of antenna in transmission and reception. This calls for in-depth knowledge and comprehension of technical adjustments and the way the components interact.

One of the most important factors in antenna optimization is the gain, which measures how an antenna concentrates energy in a particular direction. The gain is essential for applications like satellite communication and mobile networks. The higher gain results in higher strength and range signal.

For the directivity, it defines how the antenna's radiation pattern is. Like the gain, higher directivity means higher concentrations of energy in specific direction the thing that improve communication efficiency. Another crucial parameter that we consider is the bandwidth it describes the frequency range that an antenna can operate over.

The bandwidth is also crucial to data-driven because a larger bandwidth allows transmission to more data.

Efficiency is also important, this includes decreasing losses during communication, which means between transmission and reception this to making sure that maximum amount of energy is transformed into usable signals.

Always with the same principle of those parameter, more efficiency lead to the most reliable communication is. Furthermore, this field witnesses the emergence of a several innovative technologies including MIMO [multiple input multiple Output], which is a system that employs multiple antennas to enhance the performance and capacity. Those type of innovation are changing completely the domain of antenna and as a result many industries that use antenna had changed like telecommunication aerospace ...

C. Integration of Machine learning

Adopting Digital Twins as part of antenna optimization aims to develop a virtual replica of the antenna that can respond and simulate true to life behavior depending on changing conditions. With machine learning embedded, Digital Twin can inspect massive amounts of data and

anticipate performance outcomes. This provides an interesting optimization of bandwidth, directivity and gain as well. Digital Twin's capacity to adjust to evolving conditions, reduce testing time is improved by Machine learning algorithms. Positioning it as a potent tool for developing reliable antenna systems.

D. Case Study: Digital Twin in Antenna Systems Telecommunications Company Example

The benefits of the leveraging of Digital Twin technology in Antenna Systems are many:

1. Significant Reduction in Design Time

Companies such as Ericsson and Nokia have achieved considerable reductions in design time for antennas by adopting Digital Twins. For instance, Nokia were able to reduce the design time of their advanced antenna by up to 25%.

2. Enhanced Signal Strength and Coverage

The adoption of Digital Twin technology has led to noticeable improvements, particularly in terms of the antenna optimization. For instance, 5G increased by more than 25 percent for Huawei Technologies Co. since the adoption of Digital Twin technology.

3. Predictive engineering capabilities

To enable real time refinement and predict how a physical product will behave in the real world environments, a number of Digital Twin platforms have been developed for this purpose.

Among these Digital Twin platforms, we have Siemens Simcenter and Dassault Systèmes.

The capabilities that these platforms offer are significantly efficient in the sense that they guarantee optimal functionality by enabling modifications according to environmental factors.

That is why Industry leaders like Nokia, Ericsson and Huawei promote and favor innovation using Digital Twin technology in antenna optimization.

IV. COGNITIVE / ACTIVE DIGITAL TWIN

The cognitive Digital Twin presents a significant advancement in the Digital Twin field by integrating AI into the traditional Digital Twin structure.

In contrast to traditional Digital Twin that mimic the physical system, this innovation learns continuously from data then he can change and evolve over time.

The main goal of this research is to include AI in antenna systems to identify automatically performance issues, understand historical and present so as to be able to propose smart optimizations design.

For example, after integration of AI, the system must be able to analyze electromagnetic behavior and propose solutions to improve precision and solve magnetic problems like lower signal loss. As a result, the twin changes from a passive model into an intelligent smart flexible system that can adapt to environmental change and capable of making decisions on their own.

To sum up, AI is able to transform Digital Twin into an active agent with continuous improvement.

CONCLUSION

In summary, this paper examines the findings from the literature reviews of research on Digital Twin technology. It begins with a brief definition of Digital Twin, its history, and its benefits in terms of antennas optimization followed by analysis and estimation of the key differences between this emerging technology and virtualization. Additionally, we looked at Digital Model, Digital shadow vs digital twin and the primary requirements for creating them. Furthermore, we observed a few digital twin application domains. In this paper, we have attempted to show the role of AI for digital twin research and how the diverse functions of AI can be used as a key enabler of Digital Twin System for antenna to optimize performance parameters. For this reason, we finally absorbed how to optimize the antenna by integrating machine learning into the antenna's digital twin.

REFERENCES

- [1] P. Chatterjee, Y. Borole, P. Borkar, R. Raut, and V. P. Balpande, *Digital Twins: Internet of Things, Machine Learning, and Smart Manufacturing*, Percipio Book, 2021.
- [2] S. Jiang and A. Alkhateeb, "Digital Twin Based Beam Prediction: Can We Train in the Digital World and Deploy in Reality?," School of Electrical, Computer and Energy Engineering, Arizona State University, 2021. [Online]. Available: <https://arxiv.org/abs/2103.07752>
- [3] B. Salehi, U. Demir, D. Roy, S. Pradhan, J. Dy, S. Ioannidis, and K. Chowdhury, "Multiverse at the Edge: Interacting Real World and Digital Twins for Wireless Beamforming," *Proc. ACM Workshop on Wireless Network Intelligence (WiN 2022)*, New Orleans, LA, USA, 2022, pp. 27–34.
- [4] P. Chatterjee, Y. Borole, P. Borkar, R. Raut, and V. P. Balpande, *Digital Twins: Internet of Things, Machine Learning, and Smart Manufacturing*, CRC Press, 2022.
- [5] "Networked Mobile Sensing and Mobile Actuation," in *Proc. 2021 IEEE 1st Int. Conf. on Digital Twins and Parallel Intelligence (DTPi)*, Beijing, China, Jul. 15–Aug. 15, 2021, pp. 354–357. doi: 10.1109/DTPi52836.2021.00063
- [6] Microsoft, "Azure Digital Twins," [Online]. Available: <https://azure.microsoft.com/en-us/products/digital-twins> [Accessed: Jun. 5, 2024].
- [7] Amazon Web Services, "AWS IoT TwinMaker," [Online]. Available: <https://aws.amazon.com/iot-twinmaker> [Accessed: Jun. 5, 2024].
- [8] Oracle, "Oracle IoT Digital Twin Framework," [Online]. Available: <https://docs.oracle.com/en/cloud/paas/iot-cloud/iotgs/iot-digital-twin-framework.html> [Accessed: Jun. 5, 2024].
- [9] Eclipse Foundation, "Eclipse Ditto Framework for Digital Twins," [Online]. Available: <https://eclipse.dev/ditto/index.html> [Accessed: Jun. 5, 2024].

Annex 1

Criteria	Azure Digital Twins [6]	Amazon IoT TwinMaker [7]	Oracle IoT Digital Twin Framework [8]	Eclipse Ditto [9]
Framework and Modeling	Rich modeling, DTDL support	Supports AWS Cloud Formation templates	Custom models, supports PLM	JSON-based modeling
Data Integration	IoT Hub, Event Hub, Time Series Insights	IoT Core, S3, DynamoDB, Grafana	Oracle IoT Cloud Service, ERP, SCM	MQTT, HTTP, WebSockets, Kafka
Simulation and Analysis	Integrates with Azure Analytics services	Integrates with AWS analytics services	Integrates with Oracle Analytics Cloud	Limited, depends on integration
Security	Enterprise-grade security features	AWS security standards	Oracle Cloud security features	Open-source, security managed by user
Visualization Tools	Offers 2D/3D scenes visualization	Provides an immersive 3D view of systems	Integration with Oracle products	Open standard, supports various protocols
Customization	High, using Azure functions, Logic Apps	High, using Lambda, other AWS services	High, using Oracle cloud services	High, open-source allows for customization
Developer Tools	Visual Studio, Azure CLI	AWS CLI, SDKs, CloudFormation	Oracle Developer tools	Eclipse IDE, REST APIs
Community	Strong, Microsoft community	Strong, AWS community	Oracle support	Strong, open-source community